

AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 10.7: Synthesis, Exercises, and Appendix

Zhigang Feng

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Topic 10.7 Agenda

1. **Synthesis and Exercises**
2. **Appendix**

Synthesis and Exercises

Comparing reusable skills, empirical pipelines, public-web datasets, and full-lifecycle collaboration

Compare the Case-Study Workflows

Case	Artifact	Validation gate	Main failure mode
Paper review skill	referee-style report	quotes and section claims trace to the paper	generic criticism or hallucinated evidence
MEPS pipeline	panel, weights, figures	rates match external benchmarks	wrong crosswalk or dropped survey weights
ES Fellows	provenance-rich dataset	field-level source trace and missingness report	weak sources hidden by clean tables
Research collaborator	model, pipeline, critique, revision	claims survive adversarial review	polished output around weak framing
AI exposure measurement	rubric, JSON scores, comparison table, scatter	heuristic and reference comparison; invariant checks	construction choice silently drives coefficient sign
OLG homotopy sprint	V0–V5 version ladder and tests	each version inherits a validated baseline	one-shot model generation with no rollback point

When to Use Which Workflow

Build a skill

- repeated use
- high QC needs
- stable output format

Stay ad hoc

- one-off empirical job
- fast iteration
- small prompt set

Build a dataset harness

- many entities
- uneven sources
- provenance matters

Full-lifecycle collab.

- novel research question
- theory needs operationalization
- claims discipline required
- adversarial review needed

Measurement and computation cases. Use the AI-exposure lab when the core risk is construct validity; use the OLG homotopy demo when the core risk is model complexity and rollback discipline.

What Stayed Human Across All Cases

Economist	Agent
define the question and success criteria	execute bounded tasks repeatedly
set the coding and validation rules	manipulate files, prompts, and reruns
spot implausible outputs	surface drafts, diagnostics, and plots
interpret the economic meaning	accelerate the path from question to artifact

The constant: the question, identification logic, benchmark, and interpretation remain human work.

Failure Modes to Watch For

1. **Case 1:** confident but unsupported criticism
2. **Case 2:** clean-looking code with a wrong variable mapping
3. **Case 3:** a rich-looking dataset built on weak or forbidden sources
4. **Case 4:** AI validates the design instead of challenging it, or data contradicts theory without adversarial structure to surface it
5. **All cases:** hidden assumptions that never made it into project files

The cure is the same in every case: name the validation step, save the rules on disk, and inspect the output like a referee.

Exercises and Deliverables

Exercise	Artifact	Deliverable
Exercise A: MEPS lab	harmonized panel and validated figures	in-class guided run with the existing notebook
Exercise B: AI-exposure lab	rubric, scores, scatter, and validation table	run the Module 3 demo notebook; compare heuristic vs reference
Exercise C: OLG homotopy	V0->V1 transition or validation report	inspect the five-step demo; run one smoke test
Exercise D: fellows assignment	batch-level metadata dataset with provenance	normalized CSV, four figures, and a memo

- Exercise A teaches empirical execution and validation
- Exercise B teaches LLM-as-measurement and construct validation
- Exercise C teaches homotopy and rollback discipline for computational models
- Exercise D teaches controlled public-web data construction

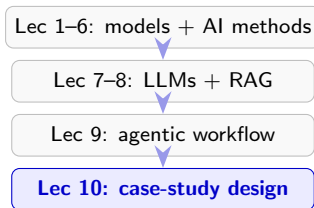
Capstone Demo Map

Demo	Where to run	Expected visible output
AI exposure	Module 3 exposure demo notebook	substitution/assistance scatter; heuristic vs reference table; coefficient-flip discussion
OLG RAG	Module 4 OLG RAG terminal demo	retrieval report; top- K evidence; grounded answer; refusal example
OLG homotopy	Module 4 OLG five-step notebook	V0–V5 version ladder; validation gates; regenerated reports

Full paths and commands are in `FullCourse_Demo_Index.md`.

Teaching rule: every demo needs one launch command, one expected artifact, and one validation question students can answer from the output.

Looking Back at the Course



Final message: the distance from question to artifact has collapsed, but economists still own the standards for what counts as evidence.

Appendix

Operational details moved out of the main narrative

Appendix A: Paper-Review Skill Internals

Component	Role
SKILL.md	orchestrates the 7-phase pipeline
shared-directives.md	confidence gate, tone, quote rules
section-prompts.md	specialized prompts by section type
editorial-prompt.md	final quality gate before output

Reason this is appendix material: students need to understand the workflow design before they need a file inventory.

Appendix B: MEPS Prompt Skeleton

Prompt A: download + parse + harmonize + save panel

Prompt B: compute weighted rates by year and subgroup

Prompt C: generate annotated publication-quality figures

Operational note: the notebook remains the handout for the in-class lab.

Appendix C: Fellows Normalized Schema

Column group	Examples
seed fields	name, affiliation, election year
source fields	official profile URL, bio URL, CV URL
biography fields	birth date raw, birth year, PhD year, gender explicit
research fields	research fields raw, normalized field bucket
audit fields	source_birth, source_phd, source_gender, source_field, notes

Appendix D: Fellows Batching Rule

- official roster parsed into 18 batches of 50
- students complete one batch at a time
- every batch must report coverage before interpretation
- if birth-year coverage stays low, use years since PhD instead of age

Appendix E: Resources

- Lecture 10 MEPS notebook in this folder
- Exercise10_ES_Fellows/ script bundle in this folder
- Official Econometric Society current fellows page, checked April 13, 2026
- MEPS public-use files and external benchmark sources used in Case Study 2